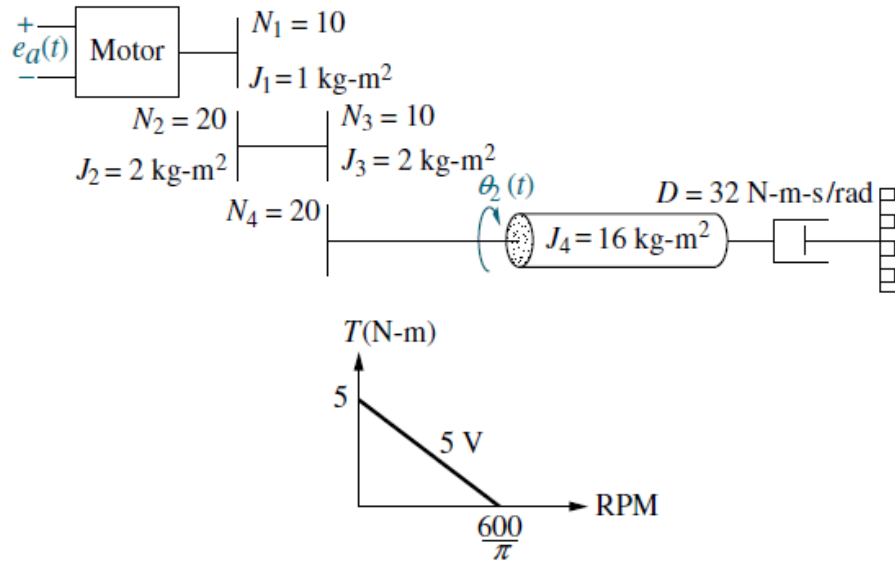


ME0344 Final Project

Q1. An armature-controlled dc motor whose torque-speed characteristics are shown below drives the load also shown.



- Find the transfer function of the system θ_m/E_a .
- Use MATLAB/Simulink to simulate the step response of the system. Mark on the plot the steady-state value, percent overshoot, and the rise time, peak time, and settling time.
- Use MATLAB/Simulink to simulate the step response of the closed-loop system assuming unity feedback. Mark on the plot the steady-state value, percent overshoot, and the rise time, peak time, and settling time. Compare the results in ii and iii.
- Design a PID controller to obtain zero-steady state error with 20% improvement on the settling time.

Q2. The open-loop transfer function of the plant for a specific application is

$$Gp(s) = \frac{6}{s(2s + 2)(3s + 24)}$$

1. Plot the root locus of the system using MATLAB.
2. Is it possible with proportional control to achieve settling time less than 2.07 sec for this plant. Explain why based on the root locus plot.
3. Use PD control to improve the response, so that the dominant time constant is 0.5 sec or less and the damping ratio is 0.707 or greater.

Project Requirements:

- i. The deadline for this 20 grades project is **Sunday 12 January 2025**.
- ii. You can work in a group (max 3 students) or individually.
- iii. You **must** submit a report with the results and simulation files (MATLAB/Simulink).